

ENVS-193DS Final

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Link to Git Hub

[GitHub Repository](#)

Set Up

```
library(tidyverse) # data wrangling and ggplot
library(here)      # file paths
library(janitor)   # data cleaning
library(DHARMA)    # diagnostic checks for models
library(MuMIn)    # AIC model selection
library(ggeffects) # model predictions
library(flextable) # making tables
```

```
nest_data <- read_csv(here("data", "nest_data_final.csv"))
```

Rows: 299 Columns: 10

-- Column specification -----

Delimiter: ","

chr (3): alive, bird_species, ant_species

dbl (7): height_cm, diameter_mm, case_control, case, canopy_width, canopy_le...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression, since the response variable (probability of American Avocet microhabitat use) is binary and distance to water edge is a continuous predictor. In part 2, they used a chi-square test of independence, since they are testing the relationship between two categorical variables (bird species and habitat type).

b. Suggestions for rewriting

Part 1: Distance to water edge was a significant predictor of American Avocet microhabitat use, meaning that where a bird was found was related to how far it was from the water's edge.

Part 2: The three bird species — American Avocet, Forster's Tern, and Black-necked Stilt — were not evenly distributed across habitat types (managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflat), suggesting that each species shows preferences for particular habitats.

c. Further analysis

Native plant percent cover (continuous, measured as the percentage of ground covered by native vegetation within a survey plot) could influence American Avocet microhabitat use because Avocets forage in shallow open water and mudflats, and higher native plant cover may reduce the availability of these open foraging areas, making those sites less suitable. This variable could be measured using standard vegetation transect surveys conducted during the breeding season at each survey plot.